

How to use Auction Cipher

Oct 8, 2026 — updated for v3.2b (one stop, Fees, Stop Placement, Break-even, Stop-Loss Lab)

Set up in five minutes

Do these once per chart. After that, the indicator tells you where the trade locations are, when a setup is complete, and what size to trade.

1. **Add it.** Open the Pine Editor, paste the code from `AuctionCipher.pine` in your GitHub repo, click *Add to chart*. It draws on the price chart and opens one lower pane.
2. **Pick a preset.** Settings → * *START HERE* → **Chart Preset**. Use **Clean** while trading, **Standard** while studying.
3. **Check the instrument.** Leave **Instrument Profile** on *Auto-detect* for BTC and gold. For anything else, set it to *Manual* and choose **Period Mode** in group 1.
4. **Enter your real numbers.** Group 25 - *LIVE RISK*: **Account size (\$)** and **Risk per trade (%)**. Start at 0.5% or less.
5. **Create one alert.** TradingView Alert → Condition: *Auction Cipher* → **Any alert()** **function call** → *Once per bar close*. Every BUY/SELL, with its size and stop, arrives through this single alert.
6. **Set your real cost per round trip.** Group 21 - *RISK LAYER*: **Fees + Spread, Round Trip (%)**. This is subtracted from every trade in the statistics and the Lab, so leaving it at 0 flatters every number you read.

Best timeframes: **1h** for intraday, **4h** for swings. On 15m and below the signals did not hold up in testing (see the last section).

Reading the chart

Three places carry everything: the **read box** at the top right of price, the **ACTION** row in the dashboard, and the **levels**. The read box and ACTION are the only two that tell you to do something; every other mark is evidence.

The two you act on

Where	What it says	Meaning
Read box, first line	BALANCE / IMBALANCE / TRANSITION + an instruction	Today's playbook (next section).
Read box	>>> BUY SIGNAL + Size ...u @ stop ...	A complete setup, with your position size and stop.
Dashboard ACTION	WAIT	Nothing to do.
Dashboard ACTION	buy setting up / sell setting up	Three gates open; the fourth is missing. Get ready, don't enter.
Dashboard ACTION	BUY 2.0R / SELL 2.0R	Signal live, with its reward-to-risk.
Dashboard ACTION	CONFLICT	Both sides qualified. Stay out.

Levels and marks on the price chart

You see	It is	Use it as
Grey lines labelled prior POC / VAH / VAL	Yesterday's value area	The main places to look for a trade.
Amber line (POC), cyan lines (VAH/VAL)	Today's developing value	Targets and context.
Pink line nPOC	An old POC price never went back to	A magnet — likely target.
Green IB lines	Initial balance (opening range)	Breakout reference.
Orange line	VWAP	Fair price now; far from it = stretched.
Yellow/red lines	Weekly POC / VAH / VAL	The bigger picture's edges.
Green BUY / red SELL label	A full four-gate signal	Your entry (next section).
Red and green boxes after a signal	Stop zone and target zone	Where you are wrong, where you get paid.

You see	It is	Use it as
× marked TP	Target reached	Trade done.
Orange square above a bar	Absorption	Someone is defending this price.
Yellow ×	Liquidation cascade	Forced selling/buying; often overshoots, then reverses.
BOS / CHoCH labels	Structure break with / against trend	CHoCH = first warning the trend may turn.
Grey boxes with a small profile	Balance area	Its edges are trade locations; a break out of it can run.

The lower pane

You see	Meaning
Dark waves crossing	WaveTrend momentum. The cross is the TRIGGER.
Green/red area near zero	Money flow — green = buyers in control.
Yellow dot / white dot	Momentum cross up / down. Big dot = a divergence came with it.
Circle / diamond on the waves	Confirmed WaveTrend / CVD divergence (marked 5 bars back — a delay, not a repaint).
Faint early marker	Divergence still forming. A warning only; it can disappear.
Strip at the bottom	Trend ribbon: green bullish, red bearish structure.

Before you trade: pick the playbook

The first line of the read box decides what kind of trade you are allowed to take today. Read it before every session and only take signals that agree with it.

Read box says	What the market is doing	Your playbook
BALANCE — Fade the edges back to POC	Two-way trade inside a range	Sell near VAH, buy near VAL, target the POC. Above value → sells favoured; below value → buys favoured; mid-value → wait for an edge.
BALANCE by value, but a TREND DAY is running — do NOT fade	Range on paper, but today keeps extending	No fading. Wait for the range to stop expanding, or trade only with the direction.
IMBALANCE — Go with it, do not fade	Value is moving to a new area	Trade with the move only. Targets: the untested POC shown in the read.
TRANSITION — Stand down	The auction hasn't decided	No trade.

The lines under the headline add detail:

- **Price is above value (expensive) / below value (cheap) / inside value** — where price sits against today's value area.
- **Zoomed out: price is ...** — the same against the weekly value area. A buy into weekly value from below is stronger than one into weekly resistance.
- **High/Low is POOR** — unfinished; price usually comes back to it. Good target, bad place to fade.
- **High/Low is EXCESS** — finished; the rejection was real. Good place to lean against.
- **ABSORPTION now / EXHAUSTION now** — a live warning on the current bar.

Pre-session checklist

- ☐ Read the headline and note the playbook.
- ☐ Note where price sits: above, below or inside value.
- ☐ Mark the two nearest trade locations: prior VAH/VAL, nPOC, weekly edge.
- ☐ Check for a POOR high or low to use as a target.
- ☐ Decide which side you are allowed to trade. Ignore signals on the other side.

Taking a BUY or SELL

A signal is only confirmed when its bar closes, so you enter on the **next bar's open**, never at the signal bar's close. The alert and the read box give you every number you need.

1. **The signal prints.** A green BUY or red SELL label appears, ACTION shows *BUY 2.0R* (or SELL), and the alert arrives: *CHART BUY ... Entry at the NEXT bar open, stop X, target Y. Stand down if the open leaves less than 1.5R. LIVE SIZE 0.1u at stop X (\$50 risk).*
2. **Check the playbook.** Does it match the read box? A buy on an IMBALANCE-down day, or any signal on TRANSITION, gets skipped.
3. **Wait for the next bar to open.** If price opened so far toward the target that less than 1.5R is left (your *Minimum R*), skip it.
4. **Enter at market** with the size from the **LIVE SIZE** line.
5. **Put your stop at the stop price (X).** The size was calculated for that one stop — the same stop the statistics and the Lab measured — so this keeps your loss at exactly your risk amount (plus the fee you set in Fees + Spread).
6. **Put your target at the target price (Y),** the top of the green box. It is capped at 2R by default.

Worked example (BTC): account \$10,000, risk 0.5% = \$50. Entry 60,000, stop 59,500 → distance \$500 → size $\$50 \div \$500 = 0.1 \text{ BTC}$. If the stop is hit you lose \$50 plus your round-trip fee, whatever the distance.

There is now **one stop**, used everywhere — the pop-up, the alert, the box on the chart, the statistics table and the Stop-Loss Lab. Which stop that is comes from **Stop Placement** (group 21); the default is *Structural (swing)*. Trade the size printed next to it. For futures, the size is in ounces or coins, so divide by the contract size (GC = 100 oz, MGC = 10 oz).

Managing and exiting the trade

Once you're in, the plan is fixed: leave the stop and target alone and let one of three things end the trade.

What happens	On the chart	What you do
Target reached	× marked TP	Take the profit at the target. Don't hold for more — in testing, trades that ran past 2R usually gave it back.

What happens	On the chart	What you do
Stop reached	Price trades through the stop	Accept the loss. It is the planned risk amount, nothing more.
A full opposite signal appears	SELL while you're long (or BUY while short)	Close at market. The reason for the trade is gone. Go flat; don't flip into the new direction.

Rules while a trade is open:

- **Don't tighten the stop**, unless you've turned on **Move Stop to Break-even at (R)** (group 21, off by default). In the backtest, stops tighter than about $1.5 \times \text{ATR}$ never reduced losses; they only got hit more often.
- **Break-even is opt-in**. Set it above 0 and, once price has moved that many R in your favour, the stop moves to your entry from the next bar on — never the same bar, since the order of a candle's high and low isn't known. A trade stopped at break-even books as FLAT, not a loss; read Expectancy, not the hit rate, once this is on. Check the Lab's *Struct + BE 1R* row first to see whether it actually helps on your chart.
- **No new signal will show** while a trade is open (*One Trade At A Time* is on). That is deliberate.
- **Track it live** in the statistics table: the *Open now (live R)* row shows how far in profit or loss the open trade is, in R.
- **CHoCH or absorption against you** near the target is a reason to take profit early, not to add to the position.

Check it works before you risk money

Auction Cipher shows you where a trade makes sense; it does not guarantee the trade wins. Our testing over the past month never proved a reliable edge. On 15m no stop setting was profitable, and on 1h the best result was a thin +\$331 over 71 trades. Prove it on your own market first.

Test it on your symbol and timeframe

1. Load as much history as TradingView allows on the chart you plan to trade.
2. Set **Slippage (ticks)** in group 21 to what your exchange really costs you.
3. Read the statistics table: **Expectancy R/trade** must be above 0, and the trade count should be at least 20 — 100+ is better. Check the *Verdict (drift-controlled)* row too.

4. Paper trade the steps above for 2–4 weeks, or about 20 trades, before using real money.
5. Change **one setting at a time**, then re-read the statistics. Never change settings because of one loss.

Rules that protect the account (suggested limits — adjust to your situation)

- Risk 0.5% or less per trade. Size always comes from the LIVE SIZE line, never by feel.
- Stop for the day after 2 losses in a row, or once the day is down 1.5%.
- One trade at a time. No revenge trades and no adding to a loser.
- No signal that disagrees with the read box. TRANSITION means no trades.
- Keep a simple log: date, signal, entry, stop, exit, R. After 20 trades, compare your R to the statistics table.

Auction Cipher — Settings Menu Guide

Oct 7, 2026 · @Zahin Farhat

How to use this guide

Open the settings at the top group, * **START HERE - CHART PRESET**: the Chart Preset there overrides most on/off switches in the rest of the menu. If you flip a toggle and nothing changes on the chart, this is why — set Chart Preset to **Custom (use the groups below)** first.

The settings groups are numbered in the menu. This guide follows them in the order they matter for trading, not menu order. Each section lists the settings worth knowing, their default, and when to change them; pure colour pickers are left out unless they matter.

The five settings to check first:

1. **Chart Preset** (* START HERE) — Clean, Standard, Everything, or Custom.
2. **Instrument Profile** (* START HERE) — leave on Auto-detect for BTC and gold.
3. **Period Mode** (1 - PERIOD) — only matters if Instrument Profile is Manual or the symbol isn't BTC/gold.
4. **Gate Mode** and **Sensitivity** (16 - CHART SIGNALS) — decide how strict BUY/SELL signals are.
5. **Account size** and **Risk per trade** (25 - LIVE RISK) — set these to your real account before reading the size in the signal pop-up.

Presets and instrument profile

Chart Preset (default Standard) decides which layers draw. The table shows what each preset actually turns on in the code. Anything not listed here is controlled by its own group's toggle.

Layer	Clean	Standard	Everything
Auction layer, prior POC/VAH/VAL, naked POCs	on	on	on
VWAP line	on	on	on
Chart BUY/SELL signals, risk layer, CME gap	on	on	on

Layer	Clean	Standard	Everything
Profile histogram, initial balance, low volume nodes	off	on	on
VWAP bands, golden pocket fibs	off	on	on
Volume DNA, DNA glow, absorption clouds	off	on	on
Market structure, balance areas, OI, liquidation cascades, statistics	off	on	on
Prior high/low, DNA circles, DNA tape, volume bubbles, fib targets, session boxes	off	off	on

Use **Clean** for live trading, **Standard** for daily analysis, **Everything** only for studying. Pick **Custom** to control every layer yourself with the toggles in each group.

Instrument Profile (default Auto-detect) reads the ticker and switches only the settings that genuinely differ by instrument. The menu still shows your typed values; the script runs on these instead:

Setting	BTC / crypto	Gold futures (GC)	Spot gold (XAUUSD)
Period Mode	24h UTC	RTH Session 08:20–13:30 New York	Chart Daily
Initial Balance	240 min	60 min	240 min
CME gap symbol	CME:BTC1!	COMEX:GC1!	COMEX:GC1!

Set it to **Manual** only for other symbols (ES, NQ, stocks, FX); then group 1 and group 20 settings apply exactly as typed.

0 - LABELS controls the text next to each line: Name Every Line (on), Label Content (Name + Price), Label Size (Small), and Show Plain-English Read (on). If labels overlap, raise *Gap Between Side-by-side Labels* (14) or *Label Collision Tolerance* (0.5× ATR).

Auction core — groups 1, 2 and 2b

These build the volume profile: POC, value area (VAH/VAL), prior levels, naked POCs and low volume nodes. Defaults are sound; the two you may touch are **Profile Rows** (if the script feels slow) and **Profile Range**.

1 - PERIOD

Setting	Default	What it does / when to change
Period Mode	24h UTC (crypto)	When a new profile starts. Overridden by Instrument Profile unless that is Manual. RTH Session for CME futures, Weekly for a composite.
RTH Session / RTH Session Timezone	08:20–13:30, New York	Used only in RTH Session mode. Not the same as the Session Ranges timezone in group 22.
Initial Balance (minutes)	240	Opening range length. 60 for a futures pit session; 240 is the same fraction of a 24h crypto day.

2 - AUCTION LAYER

Setting	Default	What it does / when to change
Enable Auction Layer	on	Master switch for the profile and its levels.
Profile Rows	250	Price precision of POC and value area. Lower to 100–150 if the chart lags.
Value Area %	70	Share of volume inside VAH–VAL. 70 is standard; leave it.
Profile Range	Auction Period	Auction Period = fixed levels for the session. Rolling Lookback = profile follows price (last N bars).
Rolling Lookback (bars)	300	Only used with Rolling Lookback. Max 480.

Setting	Default	What it does / when to change
Profile Histogram	on	Draws the profile shape on the right.
Histogram Width (%)	34	Width of the widest row as % of period bars.
Split Up / Down Volume	on	Coloured overlay per row: green = buyers dominated that price, red = sellers.
Extend To The Right / Histogram Offset	on / 8 bars	Where the histogram sits. Raise offset if it crowds price.
Prior POC / VAH / VAL	on	Yesterday's value levels — the main trade locations.
Prior High / Low	off	Previous period extremes.
Initial Balance Lines	on	IB high/low for the current period.
Naked (untested) POCs / Max / Ignore Beyond	on / 12 / 15%	Old POCs price hasn't revisited; removed once touched.
Low Volume Nodes / Threshold / Max	on / 20% of POC / 6	Thin shelves where price tends to move fast.
Excess / Poor bar counts / Extreme band	1 / 3 / range ÷ 24	Classifies the period's high and low as EXCESS (rejected) or POOR (likely revisited). Rarely needs changing.

2b - AUCTION COLORS holds the colours for the histogram, POC (amber), VAH/VAL (cyan), prior value (grey), naked POCs (pink), LVNs (purple) and IB (green).

Volume DNA and Volume Bubbles — groups 11, 11b and 12

Volume DNA labels each bar by effort (volume) versus result (bar size). **Absorption — high volume, narrow bar — is the one to watch**, especially at VAH/VAL; it also feeds the PROOF gate of the BUY/SELL signals.

DNA state	Volume	Bar spread	Meaning	Default colour
Absorption	high	narrow	Someone is soaking up all orders	orange
Effort	high	wide	Participation and follow-through agree	cyan (up) / pink (down)
Exhaustion	low	wide	Move running on empty	yellow
Stagnation	low	narrow	Nobody trading	grey (off by default)

11 - VOLUME DNA

Setting	Default	What it does / when to change
Enable Volume DNA	on	Master switch.
Volume / Spread Lookback	20 / 20	Bars used to judge "normal" volume and bar size.
Volume Std Dev Threshold	1.5	How unusual volume must be. Raise to 2.0 for fewer, stronger marks.
Spread Std Dev Threshold	1.0	How unusual bar size must be.
Mark Absorption / Effort / Exhaustion / Stagnation	on / on / on / off	Which states get marked.
Only Mark at Value Area Edges	off	Turn on to keep only DNA marks at VAH/VAL — cuts noise a lot.

11b - VOLUME DNA - VISUALS

Setting	Default	What it does
Ghost Glow Candles	on	Coloured halo around DNA bars.
Absorption Clouds / Cloud Extension	on / 10 bars	Orange zone projected forward from absorption, labelled with volume.

Setting	Default	What it does
DNA Circles with Volume	off	Circle per DNA bar, sized by how extreme volume was.
Barcode Heatmap Tape / Rows / Position	off / 20 / Middle Right	Strip showing the DNA state of the last N bars. Keep it away from the dashboard position.

12 - VOLUME BUBBLES (off by default; on in the Everything preset) marks unusually large volume clusters as Small/Medium/Big bubbles.

Setting	Default	What it does
Detection Method	Volume OR Delta	What counts as "big". Volume + Delta is strictest.
Buy/Sell Classification	Both	Colours bubbles by candle direction, delta, or both agreeing.
Small / Medium / Big Percentile	75 / 90 / 97	Size thresholds. Raise Small to 85 to show fewer bubbles.
Consensus Mode	Majority (2 of 3)	Must be big on 2 of the 3 windows (20/50/100 bars).
Show Numbers / Bubble Text	on / Volume	Text inside each bubble.

Higher-timeframe value, HTF candles, VWAP and fibs — groups 18, 18b, 13 and 14

These add slower reference levels around the session profile. The weekly value area (on by default) is the one that matters most: it tells you whether today's value sits high or low in the bigger auction.

18 - HIGHER TIMEFRAME VALUE

Setting	Default	What it does / when to change
HTF Value Area	Weekly	A second, slower profile: Off, Daily, Weekly, Monthly. Use Daily on 1–5m charts, Monthly on 4h+. Its last completed VAH/VAL also count as signal locations.
HTF POC / HTF VAH/VAL colours	yellow / red	Line colours.

18b - HTF CANDLES (off by default)

Setting	Default	What it does
Show HTF Candles	off	Draws higher-timeframe candles beside the chart, live candle included. No data request, never repaints.
Candle Timeframe	240 (4h)	Must be higher than the chart timeframe.
Candles Shown / Offset / Width / Gap	4 / 60 / 5 / 4 bars	Layout. Raise Offset if it overlaps the profile.
Extend Open/High/Low of the Live Candle	on	Rays from where the HTF open/high/low were actually made.
Label the Live Candle	on	Timeframe, % change, and where the close sits in its range (0% = low, 100% = high).

13 - VWAP

Setting	Default	What it does / when to change
Show VWAP	on	Volume-weighted average price. When VWAP and POC sit apart, value is still moving.
VWAP Anchor	Auction Period	Keeps VWAP on the same clock as the profile. Daily / Weekly / Session also available.

Setting	Default	What it does / when to change
Show Std Dev Bands / Band 1 / Band 2	on / 1.0 / 2.0	Deviation bands. Band 1 can be used as a signal location (see group 16).

14 - FIB LEVELS

Setting	Default	What it does / when to change
Golden Pockets (retracement)	on	0.5 / 0.618 / 0.786 / 0.886 of the latest swing.
Swing Window (bars)	120	Swing must be inside this window or fibs aren't drawn.
Minimum Leg Length (bars)	15	Ignores tiny swings.
Count the Golden Pocket as Confluence	off	Adds 0.618 and 0.786 to the signal confluence count. Off on purpose — it changes how many signals fire.
Custom Target Fibs (extension)	off	1.47 / 1.55 / 2.56 / 2.60 / 2.64 targets. Targets, not entries.
Hide Fibs Beyond (% from price)	25	Stops far-away extensions from squashing the price scale.

Oscillator pane — groups 3 to 8, 27 and the lower-pane visuals

The lower pane is the Market Cipher B-style momentum view. WaveTrend crosses here are the TRIGGER for chart signals; divergences are the main PROOF. Leave the WaveTrend lengths alone — every threshold in the script is tuned to them.

3 - MOMENTUM WAVES (WaveTrend)

Setting	Default	What it does
Channel / Average / Signal Length	9 / 12 / 3	The WaveTrend formula. Don't change.
Overbought 1 / 2	53 / 60	Extreme zones. A cross beyond ± 53 counts as "in the zone".

Setting	Default	What it does
Oversold 1 / 2	-53 / -60	Same, downside.

4 - MONEY FLOW

Setting	Default	What it does / when to change
Source	Volume Delta (intrabar)	Real buy-minus-sell volume. Most informative. Alternatives: Money Flow Index, Chaikin, or the OG Market Cipher formula (contains no volume).
Show Money Flow / Length / Smoothing	on / 60 / 3	The green/red area.
Auto-normalise Height / Height	on / 1.0	Keeps the area a readable size whichever source you pick.

5 - VWAP WAVE: Show VWAP Wave (on), VWAP Height (1.0) — the gap between the two WaveTrend lines.

6 - HIDDEN LINES: Show RSI (off), RSI Length (14).

7 - CVD: Show CVD Line (on), CVD Anchor (Daily — also Weekly, Session, None/rolling), Rolling Length (200).

8 - DIVERGENCES

Setting	Default	What it does / when to change
WaveTrend Divergences / CVD Divergences	on / on	Which oscillators are checked. CVD divergence = sellers pressing but not being paid.
Draw Divergence Lines / Max Lines Kept	on / 20	Lines on the pane.
Show Hidden Divergences	off	Continuation divergences, drawn as X marks.
Hidden Divergence Counts as Proof	off	Lets hidden divergences produce signals. Measure before using.

Setting	Default	What it does / when to change
Pivot Left / Right Bars	5 / 5	Right bars = confirmation delay. Markers appear 5 bars back — a delay, not a repaint.
Min / Max Bars Between Pivots	5 / 60	Range of swing spacing that counts.
Minimum Divergence Quality (0–4)	0	0 = off. Raise to 1–2 only after comparing expectancy in the statistics table.
Quality checks	Participation on, OI on, At A Real Level on, Correlated Market off	Each scores one point. Correlated Market needs a symbol (gold: TVC:DXY, Inverse on; BTC: BINANCE:ETHUSDT.P, Inverse off).

27 - FORMING DIVERGENCE shows a divergence while it is still forming, before the pivot confirms — an early warning only, never a signal. Defaults: on, Only in the Extreme Zone on, Only at an Auction Level on. It can disappear if price keeps going.

MARKET CIPHER B: VISUALS (LOWER PANE) holds the wave, money-flow and dot colours, the trend ribbon (on, between –95 and –105), and **MCB Dots Filter** (None) — set it to *Extreme Waves + Money Flow* to show only the higher-quality cross dots.

Market structure, balance areas, CME gap and sessions — groups 15, 19, 20 and 22

These draw context on the price chart. Balance-area edges also act as signal locations; the other three are visual only.

15 - MARKET STRUCTURE

Setting	Default	What it does / when to change
Show Structure (BOS / CHoCH)	on	BOS = break with the trend (continuation). CHoCH = break against it (first warning of a turn).

Setting	Default	What it does / when to change
Swing Strength	10	Bars each side to confirm a swing. Lower (5) for scalping, higher (15–20) for swings.
Label Swings (HH/HL/LH/LL)	on	Swing tags.
Max Structure Marks	12	Older marks are deleted past this.

19 - BALANCE AREAS finds ranges by volatility contraction (ADX falling) rather than by the clock, then profiles them.

Setting	Default	What it does / when to change
Detect Balance Areas	on	Master switch.
ADX Length / ADX Threshold	14 / 30	ADX falling through 30 opens a box. Higher threshold = only deep contractions.
Build Length (bars)	20	How long the box grows before it is armed for a break.
Balance Profile Bins / Draw Balance Profile	20 / on	The mini-profile inside the box.
Use Balance Edges as Signal Location	on	Box high/low count as valid BUY/SELL locations.

20 - CME GAP

Setting	Default	What it does
Show CME Gaps	on	Untraded weekend range on CME futures, drawn as a box until filled.
CME Symbol	CME:BTC1!	Auto-switched by Instrument Profile (gold uses COMEX:GC1!).
Weekend Gaps Only	on	Ignores ordinary daily-open gaps. Leave on.
Minimum Gap (%) / Max Open Gaps	0.5 / 6	Gold auto-uses a smaller minimum.

22 - SESSION RANGES (on only in the Everything preset or Custom)

Setting	Default	What it does
Show Session Ranges	on	Asia / London / New York boxes.
Asia / London / New York	00:00-08:00 / 08:00-16:00 / 13:00-21:00	Session windows.
Session Range Timezone	UTC	Timezone for these boxes only — not the RTH timezone in group 1.
Days To Keep / Label Sessions	3 / on	History and labels.

Chart BUY/SELL signals — group 16

A BUY or SELL appears on the price chart only when four layers agree, then survives a set of filters. Pane dots need momentum alone; chart signals need all of this:

1. **CONTEXT** — the regime allows the direction (no buys while value is migrating down, when Require Regime Agreement is on).
2. **LOCATION** — price touched a level and closed back: prior VAH/VAL, naked POC, initial balance, prior high/low, VWAP ± 1 band, weekly VAH/VAL, balance edges.
3. **TRIGGER** — a WaveTrend cross.
4. **PROOF** — a divergence, absorption, money flow crossing zero, OI flush, or delta divergence (whichever are switched on).
5. **Filters** — dropped if both sides qualify at once, if efficiency is too low (when on), if open interest shows real positioning against it, if the target is under the Minimum R, or if a trade is still open.

Gate Mode is the biggest switch. The code default is **Window (legacy)**: each gate only has to have happened within the last 8 bars. **Same event (strict)** requires a divergence, with the rejection and the cross belonging to that same swing — far fewer signals.

Setting	Default	What it does / when to change
Show Buy / Sell on Price Chart	on	Master switch.
Gate Mode	Window (legacy)	See above.

Setting	Default	What it does / when to change
Event Tolerance (bars)	4	Same event mode only: how far the rejection and cross may sit from the divergence pivot.
Sensitivity	Strict (all 4)	Window mode only. Standard = 3 of 4, Loose = location + trigger, Wave only = trigger alone (for testing). Seeing no signals on Window? Turn this first.
Cross Must Be In Extreme Zone	off	On = only crosses beyond ± 53 . Can mean days with no signal in a range.
Rejection Must Close In Direction	off	On = also needs an up candle for a buy. Usually leave off.
Count VWAP Bands as Location	on	VWAP ± 1 band counts as a level.
Cooldown (bars between signals)	20	Minimum gap before the same side signals again.
Trigger Window (bars)	3	How far apart cross and touch may be (Window mode).
Location Tolerance (%)	0.12	How close a wick must come to a level. Lower it if confluence is always high.
One Trade At A Time	on	No new signal while one is open.
Opposite Signal Closes Open Trade	on	A full opposite setup closes the trade (flat, not reversed).
Reversal Close Trigger	Opposite setup completes	Alternative: also needs the opposite momentum cross.
Minimum Level Confluence	1	Levels that must stack at the rejection price. 2-3 = only strong clusters, far fewer signals.
Require Efficient Price Movement / Lookback / Minimum	off / 20 / 0.25	Noise filter. Trends run 0.3-0.6; chop is under 0.15.

Setting	Default	What it does / when to change
Require Regime Agreement	on	Stops you fading a trend day. Leave on.
Regime Confirmation (closes)	3	Closes beyond prior value before a trend is recognised.
Buy / Sell Color	green / red	Signal colours.

Risk layer and Live Risk sizing — groups 21 and 25

The risk layer draws **one stop** and a target for every signal; that same stop is used for the minimum-R filter, the chart boxes, the statistics, the Stop-Loss Lab, the alert and the position size — so the trade you take is always the trade the table measured. Live Risk turns that stop into a position size in the BUY/SELL pop-up and alert. **Set Account size and Risk per trade to your real numbers before trading off the pop-up.**

21 - RISK LAYER

Setting	Default	What it does / when to change
Draw Stop / Target / R	on	Stop and target boxes. Turning it off hides the boxes only; the filter still works.

Setting	Default	What it does / when to change
Stop Placement	Structural (swing)	Which stop is the ONE stop, everywhere. Structural (swing) — beyond the lowest low / highest high of the last Stop Swing Lookback bars, plus the buffer. Setup low/high (whole signal window) — same idea, but the lookback stretches to cover the whole signal window (Trigger Window + Pivot Right Bars + 1), so the wick that defended the level can't sit outside the stop's own lookback. Wide (structural + extra ATR) — the structural stop pushed out by Widen stop past structure (group 25). Run the Stop-Loss Lab (group 28) before changing this — it measures all three, plus break-even, half-off, trailing and a time stop, on the same entries.
Stop Buffer (× ATR)	0.3	Extra room below the swing low (above the swing high for shorts).
Stop Swing Lookback	5	Bars searched for the swing the stop sits behind.
Minimum Stop Distance (× ATR)	0.6	The stop is never closer than this. See the backtest note below.
Move Stop to Break-even at (R)	0 (off)	Once a trade has moved this many R in your favour, the stop moves to entry from the next bar on. A trade stopped at break-even books as FLAT, not a loss — read Expectancy, not hit rate, with this on. Check the Lab's Struct + BE 1R row first.

Setting	Default	What it does / when to change
Fees + Spread, Round Trip (%)	0.10	Total cost of getting in AND out, as a % of price — subtracted from every trade in the statistics, the Lab, and the live position size. Binance/Bybit perps, market both ways: ~0.10. Matters most on a tight stop: a 0.4% stop with a 0.10% round trip costs 0.25R on every trade, win or lose.
Minimum R to Show Signal	1.5	Signals with target closer than 1.5× the risk are hidden.
Cap Target at (R)	2.0	Target capped at 2R. Keep it above Minimum R. 0 = no cap.
Assumed Fill	Next bar open (realistic)	Where the statistics assume you got in. Leave it.
Slippage (ticks)	0	Set to your venue's real cost so expectancy is honest.
Box Extend (bars)	25	Box length.

25 - LIVE RISK (position size)

Setting	Default	What it does
Show size in signal pop-up / alert	on	Adds a size line to the BUY/SELL pop-up and alert.
Account size (\$)	10,000	Your account.
Risk per trade (% of account)	0.5	Dollars lost if the stop is hit (plus the fee). 0.5% of \$10,000 = \$50.
Widen stop past structure (× ATR)	0.5	Only used when Stop Placement (group 21) is Wide. Pushes the stop further past the structural stop; size shrinks so dollar risk stays the same.

Size = risk dollars ÷ (distance from entry to the one stop, plus the fee). Example: \$50 risk with a \$500 stop distance and a 0.10% round-trip fee on BTC entered at 60,000 (\$60 fee) → size = \$50 ÷ \$560 ≈ **0.089 BTC**. The size is in **units of the underlying** (BTC, ounces), not contracts. For futures, divide by the contract size — GC is 100 oz, MGC is 10 oz.

28 - STOP-LOSS LAB (backtest) — Show Stop-Loss Lab (on). A table that replays every filled trade under ten stop/exit rules at once — the same entries and targets, net of your Fees + Spread setting — and reports each rule's result side by side: *LIVE (your settings)*, *Structural swing*, *Setup window*, *Wide +N ATR*, *Tight 1 ATR*, *Loose 2 ATR*, *Structural + BE 1R*, *Struct + half off 1R*, *Struct + trail 1.5 ATR*, *Struct + 24-bar time stop*. The LIVE row must match the Expectancy row in the statistics table exactly — if it doesn't, something is broken, trust neither number. Rows under 20 trades are greyed out; under about 100 trades, a 0.1R difference between rows is noise. Use this table, not guesswork, to choose Stop Placement and whether to turn on break-even.

Stop-loss backtest (BTC, Coinbase data). Sweeping *Minimum Stop Distance* from 0.25 to 6× ATR at fixed dollar risk:

- Below about 1.5× ATR the setting changes nothing — the swing-based stop is already wider.
- On **1h**, the best result was at **2.5× ATR** (28% win rate, 2.96 payoff, +\$331 over 71 trades); 1.5–2× and 3.5×+ were worse.
- On **15m**, no stop width was profitable.

So a tighter stop does not lower losses here. If you trade 1h, try **2.0–2.5** instead of 0.6. Treat it as a starting point, not proof: 71 trades is a thin sample that only just covers fees. Re-run this comparison yourself with the Stop-Loss Lab table (group 28), which now does it automatically on your own chart.

Open interest and order flow — groups 17 and 24

Open interest (perps only) tells you who is moving price; the order flow engine reads intrabar buying and selling. Both can feed the signal gates, but only OI is wired in by default.

Price	Open interest	Meaning
Up	Up	New longs — real buying
Up	Down	Shorts covering — a squeeze, tends to stall

Price	Open interest	Meaning
Down	Up	New shorts — real selling
Down	Down	Longs closing or liquidated — a flush

17 - OPEN INTEREST & FUNDING

Setting	Default	What it does / when to change
Enable Open Interest	on	Needs an OI feed (e.g. Binance BTC perp). Spot gold shows "no OI feed", which is correct.
Mark Liquidation Cascades	on	Yellow × on a wide bar where OI collapses — forced closing that often overshoots.
Count OI as Signal Proof	on	A flush into support (or a squeeze into resistance) counts as PROOF.
Veto Signals Against Real Positioning	on	Blocks a sell into rising price on rising OI, and a buy into falling price on rising OI. Keep on.
Liquidation OI Drop / Spread (std dev)	1.5 / 1.2	How extreme a bar must be to count as a cascade.
Funding Symbol (optional)	blank	Leave blank unless you know your venue's funding ticker.

24 - ORDER FLOW runs a footprint engine without drawing the grid. It changes no signal until you turn on one of the last three options.

Setting	Default	What it does / when to change
Enable Order Flow Engine	on	Reads buy/sell volume inside each bar.
Rows per Bar	14	Slices per bar. More rows = finer, more levels.
Absorption Heavy Row Multiple	2.5	How much more than an average row a price must trade to count as defended.
Absorption Edge Band (%)	30	Only the top/bottom 30% of a bar counts.

Setting	Default	What it does / when to change
Bars to Confirm / Levels Kept	3 / 6	An absorption level must hold 3 bars; it is deleted when price closes through it.
Delta Divergence Needs (% of volume)	10	Bar closes up on net selling (or the reverse) by at least this share.
Draw Absorption Levels	off	Draws the confirmed levels (orange lines).
Count Absorption as Signal Location	off	Confirmed absorption levels act as LOCATION.
Count Absorption in Confluence	off	Adds them to the confluence count.
Count Delta Divergence as Proof	off	Hidden strength/weakness counts as PROOF.

The last three are off on purpose: each one changes how many signals fire. Turn one on at a time and compare the statistics table before and after.

Dashboards, statistics and alerts — groups 10, 23 and 26

Three tables report on the chart. The main dashboard is for trading; the statistics and gate audit tables are for checking whether the signals actually work on your symbol and timeframe.

10 - DASHBOARD — Show Dashboard (on), Position (Top Right). Rows to read first:

Row	What it tells you
ACTION (top right)	What to do now: BUY, SELL, wait, or conflicting evidence.
Regime / Day type / Open	Balance or imbalance, and how the session opened.
Position / Vs prior	Where price sits against value, and today's value vs yesterday's.
POC / VAH / VAL / nPOC up/dn	Current levels and nearest untested POCs.
High / Low	EXCESS or POOR extremes for the period.

Row	What it tells you
DNA / Vol z	Current bar's Volume DNA state and how unusual its volume is.
Open int / OI z / Funding	Positioning read (perps only).
Buy gates / Sell gates	Which of the four gates are open right now, plus the live confluence count.
Balance area / ADX / Eff	Balance-box state and trend strength / efficiency.
Profile / Order flow	Which data the profile is built from, and the order-flow read.

23 - STATISTICS — Show Statistics Table (on in Standard), Outcome Horizon (200 bars), Table Position (Bottom Left). It counts every past signal on the loaded chart: win/loss/expired, hit rate, **Expectancy R per trade**, and drift-controlled long/short results. Ignore any number with fewer than about 20 trades. Expectancy above 0 after slippage is the only number that matters.

26 - GATE AUDIT — Show Gate Audit (off), Position (Middle Left). It shows how often each gate is open across the chart. If "ALL FOUR open" is high, the cooldown — not the logic — is choosing trades. Turn it on when tuning group 16.

Alerts. Create one TradingView alert on this indicator with the condition **Any alert()** **function call**. Every event then arrives through it once per bar close: CHART BUY/SELL (with the Live Risk size and stop), Grade A/B pane signals, acceptance above/below value, CVD absorption, DNA absorption/exhaustion, balance breaks, liquidation cascades and structure breaks. Forming divergences have their own two separate alert conditions.

Recommended setups and troubleshooting

Start from one of these and change only what the table lists. They are starting points, not proven edges: the BTC backtest found no profitable stop setting on 15m, and only a thin positive result on 1h.

Setting (group)	Scalping 5–15m	Intraday 1h	Swing 4h and up
Chart Preset (*)	Clean	Standard	Standard
HTF Value Area (18)	Daily	Weekly	Monthly

Setting (group)	Scalping 5–15m	Intraday 1h	Swing 4h and up
Swing Strength (15)	5	10	15–20
Sensitivity (16, Window mode)	Standard (3 of 4)	Strict (all 4)	Strict (all 4)
Cooldown (16)	10	20	20
Minimum Stop Distance (21)	1.0	2.0–2.5	2.0
Risk per trade (25)	0.25%	0.5%	0.5%
Widen stop past structure (25)	0.5	0.5	0.5

Troubleshooting

Problem	Fix
A toggle does nothing	Set Chart Preset to Custom.
Sensitivity does nothing	It only works with Gate Mode = Window. Its tooltip says Same event is the default, but the real default is Window (legacy).
No BUY/SELL for days	Sensitivity → Standard (3 of 4); Cross Must Be In Extreme Zone off; Minimum Level Confluence 1; Minimum R → 1.2.
Too many signals	Minimum Level Confluence 2; Require Efficient Price Movement on; or Gate Mode → Same event.
Chart slow or "heavy"	Profile Rows → 100–150; turn off Volume Bubbles, DNA tape and DNA circles.
Profile frozen on spot gold	Instrument Profile → Auto-detect (uses Chart Daily for spot gold).
"no OI feed" in the dashboard	Normal on spot and stocks; OI needs a perp or futures feed.
Labels overlap	Raise Gap Between Side-by-side Labels or Label Collision Tolerance (group 0).
Pop-up size looks wrong on futures	Size is in ounces/coins; divide by the contract size.

Problem	Fix
"Too many tokens" after editing the code	The script sits at TradingView's 100,256-token limit. Any added feature needs something removed.